Sri Eshwar College of Engineering
Coimbatore | Tamilnadu
An Autonomous Institution
Affiliated to Anna University, Chennai

Department of Computer Science and Engineering

Course name: Data Science

Course code: U23AD491

Title of the project :

ProcureSense: Smart Procurement Platform

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Title	:	ProcureSense: Smart Procurement Platform
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Abstract

ProcureSense: Smart Procurement Platform

- Choosing the wrong supplier leads to late deliveries, poor quality and higher costs, yet many organizations still pick suppliers using manual analysis and past reports.
- ProcureSense is a data-driven platform that studies past procurement data to predict supplier risk, estimate delivery delays and measure how reliable each supplier is.
- A supplier recommendation engine compares vendors on cost, quality, delivery performance and risk, so procurement teams can pick suitable suppliers with confidence.
- Interactive dashboards, reports and prediction history help teams monitor performance, spot risks early and find chances to save money.

Survey: Problem Addressed

Finding reliable suppliers	Hard to tell good suppliers from risky ones using manual checks.
Delivery delays	Late deliveries are noticed only after they cause problems.
Inconsistent quality	Product quality varies and is difficult to track across vendors.
Rising expenses	Procurement costs are hard to control and compare.
Too much data	Large supplier records are slow to analyze by hand.

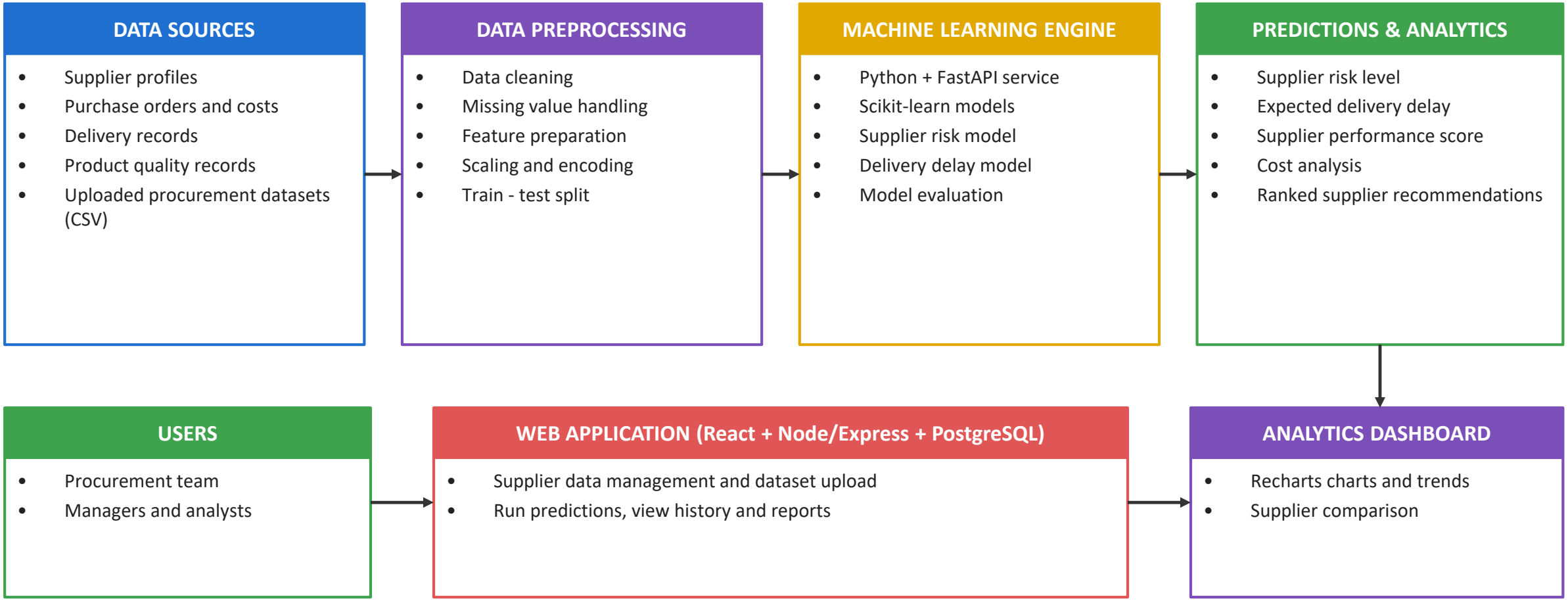
Traditional procurement systems mainly record transactions and give little predictive insight for proactive decisions.

Proposed Solution

Predict	Evaluate	Recommend	Monitor
Supplier risk and delivery delays are predicted from historical data.	Supplier performance and procurement costs are analyzed in one place.	Vendors are compared on cost, quality, delivery and risk.	Dashboards and reports show risks and cost-saving opportunities.

ProcureSense brings all of this into one centralized platform, turning procurement from reactive record-keeping into proactive decision support.

OVERALL SYSTEM ARCHITECTURE



Core Modules

1. Supplier Data Management
2. Procurement Dataset Upload and Preprocessing
3. Supplier Risk Prediction
4. Delivery Delay Prediction
5. Supplier Performance Analytics
6. Procurement Cost Analysis
7. Intelligent Supplier Recommendation Engine
8. Interactive Analytics Dashboard
9. Reports and Prediction History

Technology Stack

Layer	Technology
Frontend	React.js
Backend	Node.js and Express.js
Database	PostgreSQL
Machine Learning API	Python and FastAPI
Data Processing	Pandas and NumPy
Machine Learning	Scikit-learn
Data Visualization	Recharts

Expected Outcome

- Better supplier evaluation using clear, data-based scores
- Early identification of procurement risks
- Help with delivery planning through delay estimates
- Transparent side-by-side supplier comparisons
- Cost-saving opportunities found through data-driven recommendations

Thank You